Exercise J - A

1. Let $-\frac{\pi}{6} < \theta < -\frac{\pi}{12}$. Suppose α_1 and β_1 are the roots of the equation $x^2 - 2x\sec\theta + 1 = 0$ and α_2 and β_2 are the roots of the equation $x^2 + 2x\tan\theta - 1 = 0$. If $\alpha_1 > \beta_1$ and $\alpha_2 > \beta_2$, then $\alpha_1 + \beta_2$ equals :

[JEEAdv-2016]

माना कि $-\frac{\pi}{6} < \theta < -\frac{\pi}{12}$ है। मान लिजिये कि α_1 और β_1 समीकरण $x^2 - 2x\sec\theta + 1 = 0$ के मूल (roots) हैं और α_2 और β_2 समीकरण $x^2 + 2x\tan\theta - 1 = 0$ के मूल हैं। यदि $\alpha_1 > \beta_1$ और $\alpha_2 > \beta_2$ हैं, जब $\alpha_1 + \beta_2$ का मान है :

[JEEAdv-2016]

(A) $2(\sec\theta - \tan\theta)$ (B) $2\sec\theta$ (C) $-2\tan\theta$ (D) 0

Ans. C

Sol. Roots of the equation $x^2 - 2x\sec\theta + 1 = 0$ are $\sec\theta - \tan\theta$, $\sec\theta + \tan\theta$

$\alpha_1 = \sec\theta - \tan\theta,\ \beta_1 = \sec\theta + \tan\theta\ (\because \alpha_1 > \beta_1)$

Roots of the equation $x^2 + 2x\tan\theta - 1 = 0$ are $-\tan\theta + \sec\theta$, $-\tan\theta - \sec\theta$

$\alpha_2 = \sec\theta - \tan\theta$, $\beta_2 = -\sec\theta - \tan\theta\ (\because \alpha_2 > \beta_2)$

Hence $\alpha_1 + \beta_2 = -2\tan\theta$

2. The value of $\sum_{k=1}^{13} \frac{1}{\sin\left(\frac{\pi}{4} + \frac{(k-1)\pi}{6}\right)\sin\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)}$ is equal to : **[JEEAdv-2016]**

$\sum_{k=1}^{13} \frac{1}{\sin\left(\frac{\pi}{4} + \frac{(k-1)\pi}{6}\right)\sin\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)}$ का मान है : **[JEEAdv-2016]**

(A) $3-\sqrt{3}$ (B) $2\left(3-\sqrt{3}\right)$ (C) $2\left(\sqrt{3}-1\right)$ (D) $2\left(2+\sqrt{3}\right)$

Ans. C

Sol.
$$\sum_{k=1}^{13} \frac{\sin\left[\left(\frac{\pi}{4} + \frac{k\pi}{6}\right) - \left(\frac{\pi}{4} + (k-1)\frac{\pi}{6}\right)\right]}{\sin\frac{\pi}{6}\left(\sin\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)\sin\left(\frac{\pi}{4} + (k-1)\frac{\pi}{6}\right)\right)} = 2\sum_{k=1}^{13}\left(\cot\left(\frac{\pi}{4} + (k-1)\frac{\pi}{6}\right) - \cot\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)\right)$$

$$=2\left(\cot\frac{\pi}{4}-\cot\left(\frac{\pi}{4}+\frac{13\pi}{6}\right)\right)=2\left(1-\cot\left(\frac{29\pi}{12}\right)\right)=2\left(1-\cot\left(\frac{5\pi}{12}\right)\right)=2(1-(2-\sqrt{3}))=2(-1+\sqrt{3})$$

$$=2\left(\sqrt{3}-1\right)$$

3. Let α and β be non-zero real numbers such that $2(\cos\beta-\cos\alpha)+\cos\alpha\cos\beta=1$. Then which of the following is/are true : **[JEEAdv-2017]**

माना कि α तथा β इस प्रकार की अशून्य वास्तविक संख्यायें हैं, कि $2(\cos\beta-\cos\alpha)+\cos\alpha\cos\beta=1$. तब निम्न में से कौनसा/कौनसे कथन सत्य है/हैं : **[JEEAdv-2017]**

(A) $\tan\left(\frac{\alpha}{2}\right)+\sqrt{3}\tan\left(\frac{\beta}{2}\right)=0$ (B) $\sqrt{3}\tan\left(\frac{\alpha}{2}\right)-\tan\left(\frac{\beta}{2}\right)=0$

(C) $\tan\left(\frac{\alpha}{2}\right)-\sqrt{3}\tan\left(\frac{\beta}{2}\right)=0$ (D) $\sqrt{3}\tan\left(\frac{\alpha}{2}\right)+\tan\left(\frac{\beta}{2}\right)=0$

Ans. AC

Sol. $\cos\alpha=\left(\frac{1-a}{1+a}\right)$; $a=\tan^2\frac{\alpha}{2}$

$\cos\beta=\left(\frac{1-b}{1+b}\right)$; $b=\tan^2\frac{\beta}{2}$

$$2\left(\left(\frac{1-b}{1+b}\right)-\left(\frac{1-a}{1+a}\right)\right)+\left(\left(\frac{1-a}{1+a}\right)\left(\frac{1-b}{1+b}\right)\right)=1$$

$\Rightarrow 2((1-b)(1+a)-(1-a)(1+b))+(1-a)(1-b)=(1+a)(1+b)$
$\Rightarrow 2(1+a-b-ab-(1+b-a-ab))+1-a-b+ab=1+a+b+ab$
$\Rightarrow 4(a-b)=2(a+b)$
$\Rightarrow 2a-2b=a+b$
$\Rightarrow a=3b$

$\tan^2\frac{\alpha}{2}=3\tan^2\frac{\beta}{2}$

$\tan\frac{\alpha}{2}=\pm\sqrt{3}\tan\left(\frac{\beta}{2}\right)$

4. Let $f:[0,2]\to R$ be the function defined by $f(x)=(3-\sin(2\pi x))\sin(\pi x-\frac{\pi}{4})-\sin(3\pi x+\frac{\pi}{4})$. If $\alpha,\beta\in[0,2]$ are such that $\{x\in[0,2]: f(x)\geq 0\}=[\alpha,\beta]$, then the value of $\beta-\alpha$ is. **[JEEAdv-2020]**

माना कि फलन $f:[0,2]\to R$ को $f(x)=(3-\sin(2\pi x))\sin(\pi x-\frac{\pi}{4})-\sin(3\pi x+\frac{\pi}{4})$ द्वारा परिभाषित किया जाता है। यदि $\alpha,\beta\in[0,2]$ इस प्रकार से हैं कि $\{x\in[0,2]: f(x)\geq 0\}=[\alpha,\beta]$, तब $\beta-\alpha$ का मान है। **[JEEAdv-2020]**

Ans. 1

Sol. $f(x)=(3-\sin 2\pi x)\ \sin\left(\pi x-\frac{\pi}{4}\right)-\sin\left(3\pi x+\frac{\pi}{4}\right)$

Let $\pi x-\frac{\pi}{4}=\pi y\Rightarrow x=y+\frac{1}{4}$

$f(x)=\left(3-\sin 2\pi\left(y+\frac{1}{4}\right)\right)\sin y\,\pi-\sin\left(3\pi\left(y+\frac{1}{4}\right)+\frac{\pi}{4}\right)$

$=\left(3-\sin\left(2\pi y+\frac{\pi}{2}\right)\right)\sin y\pi-\sin\left(3\pi y+\frac{3\pi}{4}+\frac{\pi}{4}\right)$

$=(3-\cos 2\pi y)\sin\pi y+\sin(3\pi y)$

$=(3-(1-2\sin^2\pi y))\sin\pi y+3\sin\pi y-4\sin^3\pi y$

$=5\sin\pi y+2\sin^3\pi y=\sin\pi y\,(5+2\sin^2\pi y)$

Since $f(x)\geq 0\ \sin\pi y\geq 0$

$\Rightarrow \sin\pi\left(x-\frac{1}{4}\right)\geq 0\Rightarrow \pi x-\frac{\pi}{4}\in[2n\pi,(2n+1)\pi]$

$x\in[0,2]\quad\Rightarrow \pi x-\frac{\pi}{4}\in\left[-\frac{\pi}{4},\frac{7\pi}{4}\right]$

From both of these,

$\pi x-\frac{\pi}{4}\in[0,\pi]$

$\Rightarrow x\in\left[\frac{1}{4},\frac{5}{4}\right]$

So, $\alpha=\frac{1}{4},\beta=\frac{5}{4}$

$\beta-\alpha=\frac{5}{4}-\frac{1}{4}=1$

5. Let α and β be real numbers such that $-\frac{\pi}{4}<\beta<0<\alpha<\frac{\pi}{4}$. If $\sin(\alpha+\beta)=\frac{1}{3}$ and $\cos(\alpha-\beta)=\frac{2}{3}$, then the greatest integer less than or equal to $\left(\frac{\sin\alpha}{\cos\beta}+\frac{\cos\beta}{\sin\alpha}+\frac{\cos\alpha}{\sin\beta}+\frac{\sin\beta}{\cos\alpha}\right)^2$ is _____________.

[JEEAdv-2022]

माना कि α एवं β ऐसी वास्तविक संख्यायें हैं कि $-\frac{\pi}{4}<\beta<0<\alpha<\frac{\pi}{4}$ है। यदि $\sin(\alpha+\beta)=\frac{1}{3}$ एवं $\cos(\alpha-\beta)=\frac{2}{3}$ है, तब

$\left(\frac{\sin\alpha}{\cos\beta}+\frac{\cos\beta}{\sin\alpha}+\frac{\cos\alpha}{\sin\beta}+\frac{\sin\beta}{\cos\alpha}\right)^2$ से कम या बराबर महत्तम पूर्णांक _______________ है। **[JEEAdv-2022]**

Ans. 1

Sol. $\left(\frac{\sin\alpha}{\cos\beta}+\frac{\cos\alpha}{\sin\beta}+\frac{\cos\beta}{\sin\alpha}+\frac{\sin\beta}{\cos\alpha}\right)^2$

$$=\left(\frac{\cos(\alpha-\beta)}{\sin\beta\cos\beta}+\frac{\cos(\alpha-\beta)}{\sin\alpha\cdot\cos\alpha}\right)^2$$

$$=\left(\frac{4}{3}\left\{\frac{1}{\sin 2\beta}+\frac{1}{\sin 2\alpha}\right\}\right)^2$$

$$=\frac{16}{9}\left(\frac{2\sin(\alpha+\beta)\cdot\cos(\alpha-\beta)}{\sin 2\alpha\cdot\sin 2\beta}\right)^2$$

$$=\frac{16}{9}\left(\frac{4\cdot\frac{1}{3}\cdot\frac{2}{3}}{\cos(2\alpha-2\beta)-\cos(2\alpha+2\beta)}\right)^2$$

$$=\frac{16}{9}\left(\frac{\frac{8}{9}}{2\cos^2(\alpha-\beta)-1-1+2\sin^2(\alpha+\beta)}\right)^2$$

$$=\frac{16}{9}\left(\frac{\frac{8}{9}}{\frac{8}{9}-2+\frac{2}{9}}\right)^2 \qquad =\frac{16}{9}$$

6. Let $\frac{\pi}{2}<x<\pi$ be such that $\cot x=\frac{-5}{\sqrt{11}}$. Then

$\left(\sin\frac{11x}{2}\right)(\sin 6x-\cos 6x)+\left(\cos\frac{11x}{2}\right)(\sin 6x+\cos 6x)$ is equal to : **[JEE(Adv.)2024]**

माना कि $\frac{\pi}{2} < x < \pi$ इस प्रकार है कि $\cot x = \frac{-5}{\sqrt{11}}$ है। तब

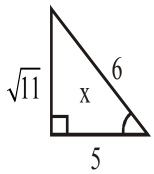
$\left(\sin\frac{11x}{2}\right)(\sin 6x - \cos 6x) + \left(\cos\frac{11x}{2}\right)(\sin 6x + \cos 6x)$ बराबर है : : **[JEE(Adv.)2024]**

(A) $\frac{\sqrt{11}-1}{2\sqrt{3}}$ (B) $\frac{\sqrt{11}+1}{2\sqrt{3}}$ (C) $\frac{\sqrt{11}+1}{3\sqrt{2}}$ (D) $\frac{\sqrt{11}-1}{3\sqrt{2}}$

Ans. B

Sol. $\frac{\pi}{2} < x < \pi$

$\cot x = -\frac{5}{\sqrt{11}}$



$$\left(\sin\frac{11x}{2}\right)(\sin 6x - \cos 6x) + \left(\cos\frac{11x}{2}\right)(\sin 6x + \cos 6x)$$

$$\left(\cos 6x \cos 11\frac{x}{2} + \sin 6x \sin 11\frac{x}{2}\right) + \left(\sin 6x \cos 11\frac{x}{2} - \cos 6x \sin 11\frac{x}{2}\right)$$

$$\Rightarrow \cos\left(6x - 11\frac{x}{2}\right) + \sin\left(6x - 11\frac{x}{2}\right) = \cos\frac{x}{2} + \sin\frac{x}{2}$$

$$\Rightarrow \sqrt{(1+\sin x)^2}$$

$$= \sqrt{\left(1+\frac{\sqrt{11}}{6}\right)^2}$$

$$\Rightarrow \sqrt{\left(1+\frac{\sqrt{11}}{6}\right)^2} = \sqrt{\frac{12+2\sqrt{11}}{12}}$$

$$= \frac{\sqrt{11}+1}{2\sqrt{3}}$$